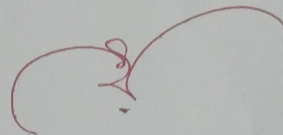
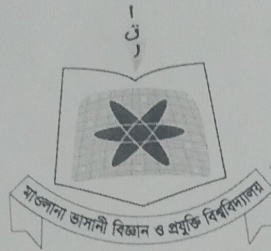


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DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No: 01

Experiment name: Comparison of Abstract Classes and Interfaces in Java
(Multiple Inheritance)

Course Title: Software Engineering and Project Management Lab

Course Code: ICT-3108

Submitted By	Submitted To
Name: Md.Imnuzzaman ID: IT23018 3 rd Year, 1 st Semester Session: 2022-23 Dept. of ICT, MBSTU	Dr. Ziaur Rahman Professor DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

Date of Performance:

Date of Submission:

Lab report : 1 compersiso between
class and interfales in terms of
inheritance in java.

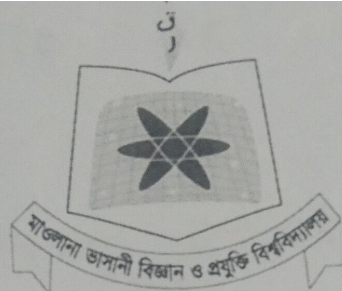
Theory:

multiple Inheritance: multiple

inheritance refers to a feature
where a class can inherit
property and behaviour from more
than one parent class java
does not support multiple
inheritance using class to avoid
ambiguity. But the support
multiple inheritance using interface.

Abstract class: An abstract class is a class that cannot be instantiated and may contain.

- abstract methods (without body)
- non-abstract methods (with body)
- Instance variables
- constructor



DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No: 02

Experiment name: Implementing Encapsulation and Data Validation in a BankAccount Class

Course Title: Software Engineering and Project Management Lab

Course Code: ICT-3108

Submitted By	Submitted To
Name: Md.Imnuzzaman ID: IT23018 3 rd Year, 1 st Semester Session: 2022-23 Dept. of ICT, MBSTU	Dr. Ziaur Rahman Professor DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

Date of Performance:

Date of Submission:

Lab report - 2 :

Lab title:

Encapsulation is an OOP concept that hides data and protect it from direct access. It answers data security by calling variable private inside a class outside class cannot change data directly. ∴ data can be changed only using public methods.

These methods check the value before saving them.

Therefore encapsulation ensure both data and security.

Java code!

```
class Account {  
    private String account number;  
    private double balance;  
    public void set account number  
        (String acc.No) {  
        if (acc.No != "" && ! acc.No.isEmpty()) {  
            account number = acc.No;  
        }  
    }  
    else {  
        System.out.println("Invalid accountnum");  
    }  
}  
    public void set Initial Balance(  
        double amount) {  
        if (amount >= 0) {  
            balance = amount;  
        }  
        else {  
            System.out.println("Balance cannot be neg!");  
        }  
    }  
}
```



```
while (re.next()) {
```

```
    System.out.println(rs.getInt(1) + "
```

```
    " + rs.getString(2)
```

```
    } }
```

```
catch (Exception e) {
```

```
    System.out.println("Error occurred");
```

```
}
```

```
finally {
```

```
    try {
```

```
        if (con != null)
```

```
            con.close();
```

```
    }
```

```
catch (Exception e) {
```

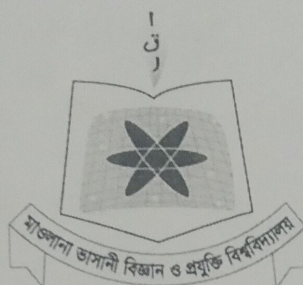
```
    System.out.println("connection not
```

```
closed");
```

```
    } } }
```


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DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No: 04

Experiment name: Database Connectivity and SELECT Query Execution Using JDBC

Course Title: Software Engineering and Project Management Lab

Course Code: ICT-3108

Submitted By	Submitted To
Name: Md.Imnuzzaman ID: IT23018 3 rd Year, 1 st Semester Session: 2022-23 Dept. of ICT, MBSTU	Dr. Ziaur Rahman Professor DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

Date of Performance:

Date of Submission:

Lab report : 4 :

In java EE application, how does a servlet controller manage the flow between the model and the view.

JDBC work as a bridge between java application and database. The linear programmer sent SQL query using JDBC. JDBC drive networks the results back to the java program. JDBC allow data to ~~the~~ be inserted, updated deleted and read. It also handles connection and error management.

Thus JDBC manages smooth connection between java and

~~while(rs.next())~~

Java code:

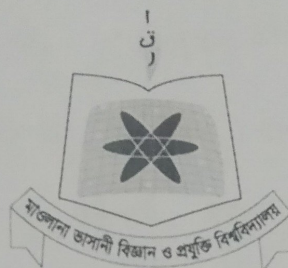
```
import java.sql.*;

class selectExample {
    public static void main (String[] args) {
        connection con = null;

        try {
            if (con != null)
                con.close();
        }
        catch (Exception e) {
            System.out.println ("connection not closed");
        }
    }
}
```


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DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No: 05

Experiment name: Servlet Controller and MVC Flow in Java EE (Forwarding Data to JSP)

Course Title: Software Engineering and Project Management Lab

Course Code: ICT-3108

Submitted By	Submitted To
Name: Md.Imnuzzaman	Dr. Ziaur Rahman
ID: IT23018	Professor
3 rd Year, 1 st Semester	DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY
Session: 2022-23	MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY
Dept. of ICT, MBSTU	

Date of Performance:

Date of Submission:

Lab Report. 5

In a java EE application, a servlet works as a controller. The controller manages the flow between model and view. The model containing business data and Logic. The servlet gets data from the model. Then it sends this data to the view. Jsp used as the view to show data to the user. The servlet forward the request to jsp using request attribute jsp reads the data and displays it thus.

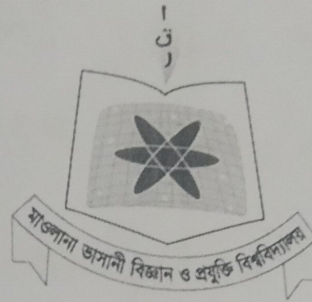
java code:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class HellowServlet extends
    HttpServlet {

    protected void doGet (HttpServlet
        request req, HttpServletResponse
        throws ServletException, IOException {
        String name = "student";
        req.setAttribute("msg", name);
        RequestDispatcher rd = req.getRequestDispatcher
            "hellow.jsp";
        rd.forward(req, res);
    }
}
```


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DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No: 06

Experiment name: Secure and Efficient Database Insert Using PreparedStatement in JDBC

Course Title: Software Engineering and Project Management Lab

Course Code: ICT-3108

Submitted By	Submitted To
Name: Md.Imnuzzaman ID: IT23018 3 rd Year, 1 st Semester Session: 2022-23 Dept. of ICT, MBSTU	Dr. Ziaur Rahman Professor DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

Date of Performance:

Date of Submission:

Lab report 6:

pre paid statement is used to execute SQL queries safely in JDBC. It improves performance because the query is precompiled by the database. The same query can be used many times with different values. Prepared statement is faster than statement for repeated queries. It also improves security by preventing SQL injection attacks. User input is treated as data, not as SQL code. Statement directing exam executes SQL and is very secure.

Java code: (Insert using prepared statement)

```
import java.sql.*;  
class insert Execute {  
    public static void main (String[]  
        range) {  
        try {  
            Connection con = DriverManager.  
                getConnection (  
                "jdbc:my sql;"  
                String sql = "Insert Insert into student  
                    (id, name) values (?)";  
            PreparedStatement ps = con.  
                prepared statement  
                    (sql)  
            ps.setInt (1, 1);  
            ps.set string (2, "Ramon");  
            ps.execute update ();  
            System.out.println ("Received insert")
```



```
con.close();
```

```
}
```

```
catch (Exception) {
```

```
system.out.println("Error occurred");
```

```
} }
```